COMPACT SIMPLE-ISOTROPY SUB-RIEMANNIAN GEODESIC ORBIT CLASSIFICATION

HUIHUI AN, ZAILI YAN, HUI ZHANG, SHAOXIANG ZHANG

Allowed standard prerequisites (editorial).

- Classical author-cited research prerequisites. Complete original embedded bibliography retained; every source-local main construction, support argument and explicitly non-adjacent continuation retained.

1. Introduction and main results

A sub-Riemannian structure on a connected smooth manifold M consists of a smooth bracket-generating subbundle $\mathcal{D} \subset TM$, together with a smooth positive definite quadratic form Q (a sub-Riemannian metric) defined on the fibers of $\mathcal{D}$. Bracket generation means that, for every $p \in M$, the local smooth sections of $\mathcal{D}$ near p, together with all their iterated Lie brackets, span the tangent space T_pM. By the Chow–Rashevskii theorem [7, 11], any two points of M can be joined by an admissible curve, that is, an absolutely continuous curve γ with $\dot\gamma(t) \in \mathcal{D}_{\gamma(t)}$ for almost every t; the sub-Riemannian distance is the infimum of the Q-lengths of admissible curves.

A geodesic is a curve that is locally a constant-speed length minimizer. By the Pontryagin Maximum Principle, length minimizers lift to extremals in the cotangent bundle T^*M. The normal extremals are precisely the integral curves of the Hamiltonian vector field of

$$H(\lambda) = \tfrac{1}{2}\, Q^*(\lambda|_{\mathcal{D}}, \lambda|_{\mathcal{D}}), \qquad \lambda \in T^*M, \tag{1}$$

where Q^* is the dual metric on the dual bundle $\mathcal{D}^*$. The projections of normal extremals are the normal geodesics. A geodesic that admits no normal lift is called strictly abnormal. It is a classical fact that the Goh condition

$$\mathcal{D}_p + [\mathcal{D}, \mathcal{D}]_p = T_pM \qquad (p \in M) \tag{2}$$

forces every geodesic to be normal; see [1, Corollary 12.14]. Here $[\mathcal{D}, \mathcal{D}]_p$ denotes the span of $[Y_1, Y_2](p)$ for local smooth sections Y_1, Y_2 of $\mathcal{D}$.

Let $(M, \mathcal{D}, Q)$ be a homogeneous sub-Riemannian manifold, and let G be a connected Lie group acting effectively and transitively on M by smooth isometries. Fix a point $p \in M$ and let H be the isotropy subgroup of G at p, so that $M = G/H$ and H is compact; denote by $\mathfrak{g}$ and $\mathfrak{h}$ the Lie algebras of G and H, respectively. Following Berestovskii [4], the structure is encoded by the following data (see [4, 13]): a reductive decomposition $\mathfrak{g} = \mathfrak{h} \oplus \mathfrak{m}$ with $\mathfrak{m}$ an $\mathrm{Ad}(H)$-invariant complement, an $\mathrm{Ad}(H)$-invariant bracket-generating subspace $\Delta \subset \mathfrak{m}$ (meaning that $\mathfrak{h} \oplus \Delta$ generates $\mathfrak{g}$ as a Lie algebra), and an $\mathrm{Ad}(H)$-invariant inner product $\langle \cdot, \cdot \rangle_\Delta$ on Δ. The distribution is $\mathcal{D} = \bigcup_{g \in G} g_*(\Delta)$

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and Q is determined by $\langle\cdot,\cdot\rangle_\Delta$. We denote this homogeneous sub-Riemannian manifold by $(G/H, \Delta, \langle\cdot,\cdot\rangle_\Delta)$.

Definition 1.1 (Sub-Riemannian geodesic orbit manifolds)**.** A connected homogeneous sub-Riemannian manifold $(M, \mathcal{D}, Q)$ is geodesic orbit (abbreviated GO) with respect to G if every normal extremal in T^*M is an orbit of a one-parameter subgroup of G under the cotangent lift of the G-action. In this case every normal geodesic is an orbit $\exp(tX)\cdot p$ of a one-parameter subgroup, and a nonzero vector $X \in \mathfrak{g}$ with this property is called a geodesic vector.

This notion generalizes the classical Riemannian geodesic orbit property [8, 9]; we refer the reader to [2, 5] for further background and recent results.

Our first main result establishes a two-step generation property; it answers in the affirmative a natural question concerning the relationship between the GO property and the Goh condition posed in [13, Remark 1.3].

Theorem 1.1 (Two-step generation)**.** *If $(M, \mathcal{D}, Q)$ is geodesic orbit with respect to G, then*

$$\mathcal{D}_p + [\mathcal{D}, \mathcal{D}]_p = T_pM \qquad \text{for every } p \in M.$$

Equivalently, for an $\mathrm{Ad}(H)$-invariant reductive decomposition $\mathfrak{g} = \mathfrak{h}\oplus\mathfrak{m}$ and the horizontal subspace $\Delta = \mathcal{D}_{eH} \subset \mathfrak{m}$,

$$\mathfrak{g} = \mathfrak{h} + \Delta + [\Delta, \Delta], \tag{3}$$

where $[\Delta, \Delta]$ denotes the linear span of the indicated Lie brackets. Consequently, every constant-speed locally length-minimizing geodesic on a sub-Riemannian geodesic orbit manifold has a normal lift; in other words, there are no strictly abnormal length minimizers.

In [6], Chen, Nikolayevsky and Nikonorov classified the compact, simply connected Riemannian geodesic orbit manifolds with simple isotropy group (see [6, Thms. 1–2]). Our second main result extends their classification to the sub-Riemannian setting.

Theorem 1.2. *Let $(M, \mathcal{D}, Q) = (G/H, \Delta, \langle\cdot,\cdot\rangle_\Delta)$ be a compact, connected, simply connected homogeneous sub-Riemannian manifold, where the action of G is almost effective and H is a connected simple Lie group. Then every proper sub-Riemannian G-GO structure belongs to one of the following two families, and every structure listed is GO and bracket generating.*

(A) *Take one of the simple-group structures in Tables 1 and 2, optionally in product with a compact simply connected semisimple group carrying any positive bi-invariant Riemannian metric.*

(B) *Let K be a compact simply connected semisimple Lie group, $H \subset K$ a connected simple subgroup that is not normal, and b any positive bi-invariant metric on K. On K take the left-invariant structure*

$$\Delta = \mathfrak{h}^{\perp_b} \subset \mathfrak{k}, \qquad Q = b|_\Delta,$$

where $\mathfrak{h}^{\perp_b}$ is the b-orthogonal complement of the Lie algebra $\mathfrak{h}$ of H in the Lie algebra $\mathfrak{k}$ of K. Its acting group is $K \times H$ by left and right translations, with isotropy group $\mathrm{diag}\, H$*. Its corank is* $\dim H$*.*

In the last case (10) of Table 2, L_x and R_x denote left and right multiplication on the real division algebra of octonions $\mathbb{O}$.

TABLE 1. Two-summand cases. In every row $\Delta = \mathfrak{m}_1$ and $A = \beta\mathrm{Id}|_{\mathfrak{m}_1}$, $\beta > 0$. The other summand is $\mathfrak{m}_2 = \mathfrak{m} \ominus \mathfrak{m}_1$. Row numbers agree with [6, Table 1].

Row	G/H	Horizontal subspace $\mathfrak{m}_1$
(1)	Spin(9)/Spin(7)	$\mathfrak{so}(9) \ominus \mathfrak{so}(8)$ (dimension 8)
(2)	Spin(10)/Spin(7)	$\mathfrak{so}(10) \ominus \mathfrak{so}(8)$ (dimension 17)
(3)	Spin(11)/Spin(7)	$\mathfrak{so}(11) \ominus \mathfrak{so}(8)$ (dimension 27)
(4)	E_6/Spin(10)	$\mathfrak{e}_6 \ominus (\mathfrak{so}(10) \oplus \mathbb{R})$ (dimension 32)
(5)	SU(n+p)/SU(n), $n \geq 2$, $1 \leq p \leq n-1$	$\mathfrak{su}(p) \oplus \mathbb{C}^{p\times n}$, where $\mathbb{C}^{p\times n}$ denotes the off-diagonal block (real dimension $p^2 - 1 + 2np$)
(6_2)	Spin($2n$+1)/SU(n), n even, $n \geq 4$	$\mathfrak{so}(2n+1) \ominus \mathfrak{so}(2n) \simeq \mathbb{C}^n$ (real dimension $2n$)
(7)	Spin($4n$+2)/SU($2n$+1), $n \geq 2$	$\mathfrak{so}(4n+2) \ominus \mathfrak{u}(2n+1)$ (dimension $2n(2n+1)$)
(8)	Sp(n+1)/Sp(n), $n \geq 1$	$\mathfrak{sp}(n+1) \ominus (\mathfrak{sp}(n) \oplus \mathfrak{sp}(1)) \simeq \mathbb{H}^n$
(11)	Spin(9)/G_2	$\mathfrak{so}(9) \ominus \mathfrak{so}(7)$ (dimension 15)

TABLE 2. The remaining simple-group cases. All parameter inequalities are part of the classification.

Row	G/H	Summands and cometric
(6_1)	Spin($2n$+1)/SU(n), n odd, $n \geq 3$	$\mathfrak{m}_1 = \mathfrak{so}(2n+1) \ominus \mathfrak{so}(2n)$, $\mathfrak{m}_2 = \mathfrak{so}(2n) \ominus \mathfrak{u}(n)$, $\mathfrak{m}_3 = \mathfrak{u}(n) \ominus \mathfrak{su}(n)$. The coefficients are $(\beta, 0, \beta/n)$, $\beta > 0$; thus $\Delta = \mathfrak{m}_1 \oplus \mathfrak{m}_3$.
(9)	SU($2n$+1)/Sp(n), $n \geq 2$	$\mathfrak{m}_1 = \mathfrak{su}(2n+1) \ominus \mathfrak{u}(2n)$, $\mathfrak{m}_2 = \mathfrak{su}(2n) \ominus \mathfrak{sp}(n)$, $\mathfrak{m}_3 = \mathfrak{u}(2n) \ominus \mathfrak{su}(2n)$. The coefficients satisfy $\beta_1 > 0$, $\beta_2, \beta_3 \geq 0$, $\beta_2\beta_3 = 0$.
(10)	Spin(8)/G_2	Identify $\mathfrak{so}(8) = \mathrm{Der}(\mathbb{O}) \oplus L_{\mathrm{Im}\,\mathbb{O}} \oplus R_{\mathrm{Im}\,\mathbb{O}}$ and put $V_{a,b} = \{aL_x + bR_x : x \in \mathrm{Im}\,\mathbb{O}\}$. Then $A = \beta\mathrm{Id}|_{V_{a,b}}$, $\beta > 0$, where $[a : b] \in \mathbb{R}P^1$ and $(a+b)(2a-b)(a-2b) \neq 0$.

2. The geodesic lemma and the proof of Theorem 1.1

Throughout this section, $(G/H, \Delta, \langle\cdot,\cdot\rangle_\Delta)$ is a connected homogeneous sub-Riemannian manifold. Since H is compact, $\langle\cdot,\cdot\rangle_\Delta$ extends to an $\mathrm{Ad}(H)$-invariant inner product $\langle\cdot,\cdot\rangle$

on $\mathfrak{m}$; such an extension is called a Riemannian extension of $\langle\cdot,\cdot\rangle_\Delta$, and $\Delta^\perp$ denotes the $\langle\cdot,\cdot\rangle$-orthogonal complement of Δ in $\mathfrak{m}$. For $X, Y \in \mathfrak{g}$ we write $[X,Y]_\mathfrak{m}$ for the projection of the Lie bracket $[X,Y]$ to $\mathfrak{m}$ along $\mathfrak{h}$. We now recall the geodesic lemma and its covector form. The cotangent-lift formulation below follows Tóth [12] and Podobryaev [10], and was cast in the following algebraic form by Yan–An–Deng [13]; it is the sub-Riemannian analogue of the Kowalski–Vanhecke geodesic lemma [9].

Lemma 2.1 (Geodesic lemma)**.** *Let $(G/H, \Delta, \langle\cdot,\cdot\rangle_\Delta)$ be a connected homogeneous sub-Riemannian manifold, let $\langle\cdot,\cdot\rangle$ be an $\mathrm{Ad}(H)$-invariant Riemannian extension of $\langle\cdot,\cdot\rangle_\Delta$ on $\mathfrak{m}$, and let $\Delta^\perp$ be the $\langle\cdot,\cdot\rangle$-orthogonal complement of Δ in $\mathfrak{m}$. Then $(G/H, \Delta, \langle\cdot,\cdot\rangle_\Delta)$ is geodesic orbit with respect to G if and only if, for every $X \in \Delta$ and every $Y \in \Delta^\perp$, there exists $Z \in \mathfrak{h}$ such that*

$$\langle X+Y, [X+Z, W]_\mathfrak{m}\rangle = 0 \qquad \text{for all } W \in \mathfrak{g}. \tag{4}$$

In this case the normal geodesic through the base point eH is of the form $\exp t(X+Z)\cdot eH$, $t \in \mathbb{R}$.

We next present a covector reformulation of (4), which is better suited to the proofs in Section 3. Let

$$\mathfrak{h}^0 := \{\lambda \in \mathfrak{g}^* : \lambda|_\mathfrak{h} = 0\}$$

be the annihilator of $\mathfrak{h}$ in $\mathfrak{g}^*$, identified with the cotangent space $T^*_{eH}(G/H)$ at the base point. Define the linear cometric map $\mathcal{A} : \mathfrak{h}^0 \to \Delta$ by

$$Q(\mathcal{A}\lambda, X) = \lambda(X), \qquad X \in \Delta, \tag{5}$$

so that $A\lambda$ is the horizontal velocity of the normal extremal with initial covector λ. The Hamiltonian (1) restricts to the quadratic form

$$H(\lambda) = \tfrac{1}{2}\,\lambda(\mathcal{A}\lambda) = \tfrac{1}{2}\,Q(\mathcal{A}\lambda, \mathcal{A}\lambda), \qquad \lambda \in \mathfrak{h}^0.$$

Lemma 2.2. *$(G/H, \Delta, \langle\cdot,\cdot\rangle_\Delta)$ is geodesic orbit with respect to G if and only if, for every $\lambda \in \mathfrak{h}^0$, there exists $Z \in \mathfrak{h}$ such that*

$$\lambda\big([\mathcal{A}\lambda + Z, W]\big) = 0 \qquad \text{for all } W \in \mathfrak{g}. \tag{6}$$

Proof. Writing $\lambda \in \mathfrak{h}^0$ as $\lambda = \langle X+Y, \cdot\rangle$ with $X = \mathcal{A}\lambda \in \Delta$ its horizontal part and $Y \in \Delta^\perp$ its vertical part, equation (4) is equivalent to (6). □

Proof of Theorem 1.1. Since the sub-Riemannian structure is homogeneous, it suffices to verify (3) at the base point $p = eH$; the passage from $\mathfrak{g} = \mathfrak{h}+\Delta+[\Delta,\Delta]$ to $\mathcal{D}_p + [\mathcal{D},\mathcal{D}]_p = T_pM$ for arbitrary p is the homogeneous-distribution correspondence of Berestovskii [4], followed by translation by G.

Let $\pi : \mathfrak{g} \to \Delta^\perp$ be the natural projection along $\mathfrak{h} \oplus \Delta$. Define

$$V := \mathrm{span}\{\pi([X_1, X_2]) : X_1, X_2 \in \Delta\}, \qquad U := V^\perp \cap \Delta^\perp.$$

Both V and U are $\mathrm{Ad}(H)$-invariant. We claim that

$$[\Delta, V] \subset \mathfrak{h} \oplus \Delta \oplus V. \tag{7}$$

Fix $X \in \Delta$, $v \in V$ and $u \in U$. By Lemma 2.1, there exists $Z \in \mathfrak{h}$ such that

$$\langle X+u, [X+Z, W]_\mathfrak{m}\rangle = 0 \qquad \forall\, W \in \mathfrak{g}.$$

Taking $W = v$ gives

$$0 = \langle X + u, [X + Z, v]_{\mathfrak{m}} \rangle = \langle X + u, [X, v]_{\mathfrak{m}} \rangle + \langle X + u, [Z, v]_{\mathfrak{m}} \rangle.$$

Since $[\mathfrak{h}, V] \subset V \subset \mathfrak{m}$, we have $[Z, v] = [Z, v]_{\mathfrak{m}} \in V$. Hence $\langle X + u, [Z, v]_{\mathfrak{m}} \rangle = 0$, and we obtain

$$\langle X + u, [X, v]_{\mathfrak{m}} \rangle = 0. \tag{8}$$

This identity holds for every $u \in U$, including $u = 0$. Subtracting the case $u = 0$ from (8) yields

$$\langle u, [X, v]_{\mathfrak{m}} \rangle = 0, \qquad \forall\, u \in U.$$

Thus $[X, v]_{\mathfrak{m}}$ has no U-component. Since $\mathfrak{m} = \Delta \oplus \Delta^{\perp} = \Delta \oplus V \oplus U$, this means $[X, v]_{\mathfrak{m}} \in \Delta \oplus V$, and therefore $[X, v] \in \mathfrak{h} \oplus \Delta \oplus V$, proving (7). Set

$$F := \mathfrak{h} \oplus \Delta \oplus V.$$

By construction,

$$F = \mathfrak{h} + \Delta + [\Delta, \Delta]. \tag{9}$$

Indeed, $[\Delta, \Delta] \subset \mathfrak{h} \oplus \Delta \oplus V = F$ and $\mathfrak{h} + \Delta + [\Delta, \Delta] \subset F$, while conversely $V \subset [\Delta, \Delta] + \mathfrak{h} + \Delta$, so that $F \subset \mathfrak{h} + \Delta + [\Delta, \Delta]$. Note also that

$$[\mathfrak{h} \oplus \Delta, F] \subset F. \tag{10}$$

Now F contains $\mathfrak{h} \oplus \Delta$, and we prove that F contains every iterated Lie bracket of elements of $\mathfrak{h} \oplus \Delta$. By the Jacobi identity and antisymmetry, every such iterated Lie bracket is a linear combination of left-nested brackets $[y_1, [y_2, [\cdots, [y_{k-1}, y_k] \cdots]]]$ with $y_i \in \mathfrak{h} \oplus \Delta$. We argue by induction on k. For $k = 1$ we have $y_1 \in \mathfrak{h} \oplus \Delta \subset F$. For $k \geq 2$, the induction hypothesis gives $[y_2, [\cdots, [y_{k-1}, y_k] \cdots]] \in F$, and since $y_1 \in \mathfrak{h} \oplus \Delta$, equation (10) gives $[y_1, [y_2, [\cdots]]] \in F$. Hence all iterated Lie brackets lie in F, and so $F = \mathfrak{g}$. By (9) we obtain $\mathfrak{g} = \mathfrak{h} + \Delta + [\Delta, \Delta]$, which is (3). The final assertion follows from the Goh condition (2). □

We conclude this section by describing the nilradical $\mathfrak{n}$ of $\mathfrak{g}$.

Lemma 2.3. *Suppose that $(G/H, \Delta, \langle \cdot, \cdot \rangle_{\Delta})$ is geodesic orbit with respect to G. Every abelian ideal $\mathfrak{a}$ of $\mathfrak{g}$ has zero intersection with $\mathfrak{h}$. Every $\mathrm{Ad}(H)$-invariant positive definite inner product on $\mathfrak{a}$ is $\mathrm{ad}(\mathfrak{g})$-invariant. In particular, $[\mathfrak{n}, \mathfrak{a}] = 0$.*

Proof. For $X \in \mathfrak{a}$, one has $(\operatorname{ad} X)^2 = 0$ because $\mathfrak{a}$ is an abelian ideal. If also $X \in \mathfrak{h}$, then the compactness of H provides an $\mathrm{Ad}(H)$-invariant positive definite inner product on $\mathfrak{g}$ for which $\operatorname{ad} X$ is skew-adjoint. Hence $\operatorname{ad} X$ is both nilpotent and skew-adjoint, so $\operatorname{ad} X = 0$. Effectiveness of the G-action implies $\mathfrak{h} \cap \mathfrak{z}(\mathfrak{g}) = 0$, where $\mathfrak{z}(\mathfrak{g})$ is the center, so $X = 0$. This proves $\mathfrak{a} \cap \mathfrak{h} = 0$.

Fix an $\mathrm{Ad}(H)$-invariant inner product b on $\mathfrak{a}$. Take $v \in \mathfrak{a}$ and let $\mu = b(v, \cdot) \in \mathfrak{a}^*$. Since $\mathfrak{a} \cap \mathfrak{h} = 0$, the restriction map $\mathfrak{h}^0 \to \mathfrak{a}^*$ is surjective. Hence we may choose $\lambda_0 \in \mathfrak{h}^0$ extending μ, that is, $\lambda_0|_{\mathfrak{a}} = \mu$. Define

$$\eta(X) = \mu([X, v]) \ (X \in \mathfrak{g}), \qquad \lambda_t = \lambda_0 + t\eta \ (t \in \mathbb{R}).$$

The ideal $\mathfrak{a}$ is abelian, so $\eta|_{\mathfrak{a}} = 0$. Also $\eta|_{\mathfrak{h}} = 0$, since b is $\mathrm{Ad}(H)$-invariant. Thus $\eta \in \mathfrak{h}^0$ and $\lambda_t \in \mathfrak{h}^0$ with $\lambda_t|_{\mathfrak{a}} = \mu$ for all t.

Apply the covector form (6) to λ_t, there exists $Z \in \mathfrak{h}$ such that

$$0 = \lambda_t\big([\mathcal{A}\lambda_t + Z, v]\big).$$

Both $[\mathcal{A}\lambda_t, v]$ and $[Z, v]$ lie in the ideal $\mathfrak{a}$. Hence $\lambda_t([Z, v]) = \mu([Z, v]) = b(v, [Z, v]) = 0$. Consequently

$$0 = \lambda_t\big([\mathcal{A}\lambda_t, v]\big). \tag{11}$$

Now $[\mathcal{A}\lambda_t, v] \in \mathfrak{a}$ and $\lambda_0|_{\mathfrak{a}} = \mu$, so $\lambda_0([\mathcal{A}\lambda_t, v]) = \mu([\mathcal{A}\lambda_t, v]) = \eta(\mathcal{A}\lambda_t)$. Also $\eta([\mathcal{A}\lambda_t, v]) = \mu([[\mathcal{A}\lambda_t, v], v]) = 0$ because $(\operatorname{ad} v)^2 = 0$ on $\mathfrak{a}$. Using $\lambda_t = \lambda_0 + t\eta$ and the linearity of $\mathcal{A}$, equation (11) gives

$$0 = \lambda_t\big([\mathcal{A}\lambda_t, v]\big) = \eta(\mathcal{A}\lambda_t) = \eta(\mathcal{A}\lambda_0) + t\,\eta(\mathcal{A}\eta) \quad (t \in \mathbb{R}).$$

Taking $t = 0$ yields $\eta(\mathcal{A}\lambda_0) = 0$, and then $t\,\eta(\mathcal{A}\eta) = 0$ for all t, so $\eta(\mathcal{A}\eta) = 0$. But $\mathcal{A}\eta \in \Delta$ is the horizontal velocity of the covector η, and by (5),

$$\eta(\mathcal{A}\eta) = Q(\mathcal{A}\eta, \mathcal{A}\eta) = Q^*(\eta|_\Delta, \eta|_\Delta) = 0.$$

Thus $\eta|_\Delta = 0$, that is, $b(v, [X, v]) = \eta(X) = 0$ for all $X \in \Delta$ and $v \in \mathfrak{a}$. Polarizing in v, we get $b(v, [X, w]) + b(w, [X, v]) = 0$, so $\operatorname{ad} X|_{\mathfrak{a}}$ is skew-adjoint with respect to b for every $X \in \Delta$. The same holds for $X \in \mathfrak{h}$ by the $\mathrm{Ad}(H)$-invariance of b. Since $\mathfrak{h} + \Delta$ generates $\mathfrak{g}$ and the skew-adjoint endomorphisms of $\mathfrak{a}$ form a Lie algebra, it follows that $\operatorname{ad} X|_{\mathfrak{a}}$ is skew-adjoint for all $X \in \mathfrak{g}$; that is, b is $\mathrm{ad}(\mathfrak{g})$-invariant.

Finally, if $X \in \mathfrak{n}$, then $\operatorname{ad} X$ on $\mathfrak{g}$ is nilpotent. Its restriction to $\mathfrak{a}$ is both nilpotent and skew-adjoint, hence zero. Thus $[\mathfrak{n}, \mathfrak{a}] = 0$. □

Theorem 2.4. *Suppose that $(G/H, \Delta, \langle\cdot,\cdot\rangle_\Delta)$ is geodesic orbit with respect to G. Then $[\mathfrak{n}, [\mathfrak{n}, \mathfrak{n}]] = 0$.*

Proof. Write the lower central series $C^1\mathfrak{n} = \mathfrak{n}$, $C^{j+1}\mathfrak{n} = [\mathfrak{n}, C^j\mathfrak{n}]$. Suppose the nilpotency class were $r \geq 3$, that is, $C^{r+1}\mathfrak{n} = 0$ but $C^r\mathfrak{n} \neq 0$. Then the ideal $\mathfrak{a} = C^{r-1}\mathfrak{n}$ would be abelian, since

$$[C^{r-1}\mathfrak{n}, C^{r-1}\mathfrak{n}] \subset C^{2r-2}\mathfrak{n} = 0.$$

It is an ideal of $\mathfrak{g}$ because the terms of the lower central series are characteristic in $\mathfrak{n}$ and $\mathfrak{n}$ is an ideal of $\mathfrak{g}$. Lemma 2.3 gives $[\mathfrak{n}, \mathfrak{a}] = 0$, contradicting $[\mathfrak{n}, C^{r-1}\mathfrak{n}] = C^r\mathfrak{n} \neq 0$. Hence $r \leq 2$, that is, $[\mathfrak{n}, [\mathfrak{n}, \mathfrak{n}]] = 0$. □

The following example shows that $\Delta \cap \mathfrak{n}$ may vanish.

Example 2.5. Let $\mathfrak{g} = \mathfrak{su}(2) \oplus \mathbb{R}T \oplus \mathbb{R}S$, with $\mathfrak{su}(2) = \operatorname{span}\{E_1, E_2, K\}$ and cyclic bracket $[E_1, E_2] = K$. Make these five vectors orthonormal for a bi-invariant inner product B, and set

$$\mathfrak{h} = \mathbb{R}(K + T), \qquad V = K - T, \qquad L = V + S, \qquad W = V - 2S,$$

$$\Delta = \operatorname{span}\{E_1, E_2, L\}, \qquad Q = B|_\Delta.$$

This is a compact homogeneous model $(\mathrm{SU}(2) \times T^2)/S^1$, with compatible periods for the diagonal circle and a finite ineffective kernel removed. In $\mathfrak{m} = \mathfrak{h}^{\perp_B}$, write $\xi = X + aL + bW$ with $X \in \operatorname{span}\{E_1, E_2\}$. Its cometric image is $X + aL$, and a direct computation gives

$$[X + aL + b(K + T), \xi] = 0,$$

so the structure is GO. The bracket $[E_1, E_2] = K$ also shows $\mathfrak{h} + \Delta + [\Delta, \Delta] = \mathfrak{g}$. But

$$\Delta \cap \mathfrak{su}(2) = \operatorname{span}\{E_1, E_2\}, \qquad \Delta \cap \mathfrak{n} = 0,$$

so their sum is smaller than Δ: the horizontal space need not split along the Levi decomposition.

3. Compact cometrics and the proof of Theorem 1.2

In this section we suppose that G is compact semisimple. Fix a positive definite $\mathrm{Ad}(G)$-invariant inner product B on $\mathfrak{g}$ and put $\mathfrak{m} = \mathfrak{h}^{\perp_B}$. An invariant horizontal metric is equivalently specified by an H-equivariant, B-self-adjoint, nonnegative operator $A : \mathfrak{m} \to \mathfrak{m}$ satisfying

$$\Delta = \operatorname{Im} A, \qquad Q(Ap, Aq) = B(p, Aq).$$

This is well defined on the image and positive definite there. Conversely, every invariant sub-Riemannian structure (Δ, Q) determines one such A (extend Q by zero on the orthogonal complement $\Delta^\perp$). The operator must additionally satisfy bracket generation.

Lemma 3.1 (Regularization)**.** *The cotangent GO condition is*

$$\text{for every } p \in \mathfrak{m} \text{ there is } Z \in \mathfrak{h} \text{ such that } [p, Ap + Z] = 0. \tag{12}$$

For every $t > 0$, this holds for A if and only if it holds for $A + tI$. Hence (Δ, Q) is GO if and only if the positive definite Riemannian metric with metric operator $(A + tI)^{-1}$ is G-GO.

Proof. Represent $\lambda \in \mathfrak{h}^0$ by $p \in \mathfrak{m}$ using B, that is, $\lambda = B(p, \cdot)$. Then by the ad-invariance of B, we obtain for all $W \in \mathfrak{g}$

$$\lambda([A\lambda + Z, W]) = B(p, [Ap + Z, W]) = B([p, Ap + Z], W).$$

It follows that (6) is equivalent to (12), since B is nondegenerate.

Finally,

$$[p, (A + tI)p + Z] = [p, Ap + t\,p + Z] = [p, Ap + Z] + t\,[p, p] = [p, Ap + Z],$$

so the same isotropy vector Z works on both sides for every real t. Since $A + tI$ is positive definite for $t > 0$, it defines a Riemannian metric (with cometric $A + tI$ and metric operator $(A + tI)^{-1}$) whose GO condition is exactly (12) for $A + tI$. □

Remark 3.2. Regularization changes the horizontal metric, so it is not a Riemannian extension in the sense of Section 2.

We fix the following notation. Write $\mathfrak{g} = \bigoplus_{i=1}^{N} \mathfrak{g}_i$ as a sum of compact simple ideals and let $\rho_i : \mathfrak{h} \to \mathfrak{g}_i$ be the isotropy projection. Partition the indices into

$$I = \{i : \rho_i = 0\}, \qquad P = \{i : 0 \neq \rho_i(\mathfrak{h}) \subsetneq \mathfrak{g}_i\}, \qquad F = \{i : \rho_i(\mathfrak{h}) = \mathfrak{g}_i\},$$

and put $J = P \cup F$ and $r = |J|$. We call I the inactive set, P the proper active set, and F the full set. Each active ρ_i is injective, hence an isomorphism onto its image. Let $B_\mathfrak{h}$ be the negative Killing form of $\mathfrak{h}$ and choose positive invariant forms B_i on $\mathfrak{g}_i$ such that $\rho_i^* B_i = B_\mathfrak{h}$ for $i \in J$; on inactive factors choose any positive invariant normalization. Define

$$B = \sum_i B_i, \qquad \mathfrak{m} = \mathfrak{h}^{\perp_B}, \qquad C|_{\mathfrak{g}_i} = c_i \mathrm{Id}, \qquad A_c = P_\mathfrak{m}(C|_\mathfrak{m}). \tag{13}$$

Inactive factors are full-rank Riemannian factors carrying bi-invariant metrics, since a left-invariant metric on a compact semisimple Lie group is geodesic orbit need to be bi-invariant; such factors therefore split off as direct product factors. Hence we may assume that $I = \varnothing$ without loss of generality. There are two cases: $r = 1$ and $r \geq 2$, treated in Sections 3.1 and 3.2, respectively.

3.1. **The simple case: proof of Theorem 1.2(A).** Here $r = 1$, so $\mathfrak{g}$ is simple. By Lemma 3.1, for every $t > 0$ the positive definite operator $A + tI$ defines a G-GO Riemannian metric μ_t on G/H, with cometric $A + tI$ and metric operator $(A + tI)^{-1}$. By the Riemannian classification of Chen–Nikolayevsky–Nikonorov [6, Thms. 1–2], a G-GO Riemannian metric on G/H with $\mathfrak{g}$ and $\mathfrak{h}$ simple is either naturally reductive or belongs to the exceptional (non-naturally-reductive) list of [6, Table 1]. Since naturally reductive metrics on G/H are unique up to scale, a proper G-GO structure must come from the exceptional list.

We now fix a Lie algebra pair $(\mathfrak{g}, \mathfrak{h})$ from [6, Table 1], together with a B-orthogonal decomposition

$$\mathfrak{g} = \mathfrak{h} \oplus \mathfrak{m}_1 \oplus \cdots \oplus \mathfrak{m}_s$$

into irreducible $\mathfrak{h}$-submodules $\mathfrak{m}_i$ of $\mathfrak{m}$, and write the cometric as $A = \bigoplus_i \beta_i \mathrm{Id}|_{\mathfrak{m}_i}$ with $\beta_i \geq 0$. The horizontal space is then $\Delta = \bigoplus_{\beta_i > 0} \mathfrak{m}_i$, and bracket generation means $\mathfrak{h} + \Delta + [\Delta, \Delta] = \mathfrak{g}$. By Lemma 3.1, the sub-Riemannian metric with cometric A is GO if and only if, for some $t > 0$, the positive invariant Riemannian metric $\sum_i \alpha_i B|_{\mathfrak{m}_i}$ on G/H is GO, where $\alpha_i = (\beta_i + t)^{-1}$. We check these conditions case by case.

Case 1: the two-summand rows of Table 1. Suppose $s = 2$, so $\mathfrak{m} = \mathfrak{m}_1 \oplus \mathfrak{m}_2$. In every two-summand row of [6, Table 1] the positive invariant GO metrics form the full two-parameter family: both coefficients α_1, α_2 are arbitrary positive numbers, and every positive invariant metric is GO. On the boundary we may set $\beta_1 = 0$ or $\beta_2 = 0$.

If $\beta_1 = 0$ and $\beta_2 > 0$, then $\Delta = \mathfrak{m}_2$. In each row $\mathfrak{h} \oplus \mathfrak{m}_2$ is the proper intermediate subalgebra indicated in [6, Table 1]: for example, in row 1 one has $\mathfrak{h} \oplus \mathfrak{m}_2 = \mathfrak{so}(8) \subsetneq \mathfrak{so}(9)$, and the same pattern holds in every row. Hence $\mathfrak{h} + \Delta + [\Delta, \Delta] \subset \mathfrak{h} \oplus \mathfrak{m}_2 \subsetneq \mathfrak{g}$, and bracket generation fails.

If $\beta_2 = 0$ and $\beta_1 > 0$, then $\Delta = \mathfrak{m}_1$, and we claim that Δ is bracket generating. Set $\mathfrak{l} = \mathfrak{h} \oplus \mathfrak{m}_2$; this is a proper subalgebra of $\mathfrak{g}$, and $\mathfrak{m}_1 = \mathfrak{l}^{\perp_B}$. Since B is ad-invariant, for $u \in \mathfrak{l}$ and $v, w \in \mathfrak{m}_1$ we have

$$B([u, v], w) = -B(v, [u, w]) = 0,$$

because $[u, w] \in \mathfrak{l}$ and $v \perp \mathfrak{l}$; hence $[\mathfrak{l}, \mathfrak{m}_1] \subset \mathfrak{m}_1$, that is, $\mathfrak{m}_1$ is $\mathfrak{l}$-stable. Let $\mathfrak{a}$ be the Lie algebra generated by $\mathfrak{m}_1$. Since the adjoint action of $\mathfrak{l}$ is by derivations and stabilizes the generators, $[\mathfrak{l}, \mathfrak{a}] \subset \mathfrak{a}$; and since $\mathfrak{a}$ is a subalgebra containing $\mathfrak{m}_1$, also $[\mathfrak{m}_1, \mathfrak{a}] \subset \mathfrak{a}$. Writing $\mathfrak{g} = \mathfrak{l} \oplus \mathfrak{m}_1$, we conclude $[\mathfrak{g}, \mathfrak{a}] \subset \mathfrak{a}$: the algebra $\mathfrak{a}$ is a nonzero ideal of the simple algebra $\mathfrak{g}$, hence $\mathfrak{a} = \mathfrak{g}$. Thus $\mathfrak{h} \oplus \mathfrak{m}_1$ generates $\mathfrak{g}$, that is, the structure with $\Delta = \mathfrak{m}_1$ and $A = \beta_1 \mathrm{Id}|_{\mathfrak{m}_1}$, $\beta_1 > 0$, is bracket generating and GO. This proves that the two-summand rows contribute exactly the structures of Table 1, and nothing else.

Case 2: row (6_1) with n odd. Here $G/H = \mathrm{Spin}(2n+1)/\mathrm{SU}(n)$, $n \geq 3$ odd, with $\mathfrak{h} = \mathfrak{su}(n)$ and

$$\mathfrak{m}_1 = \mathfrak{so}(2n{+}1) \ominus \mathfrak{so}(2n), \qquad \mathfrak{m}_2 = \mathfrak{so}(2n) \ominus \mathfrak{u}(n), \qquad \mathfrak{m}_3 = \mathfrak{u}(n) \ominus \mathfrak{su}(n),$$

three summands, all contained in $\mathfrak{so}(2n)$ except $\mathfrak{m}_1$. The GO condition for the positive metrics of this row, expressed in cometric coefficients, is the single relation

$$n\beta_3 = (n-1)\beta_2 + \beta_1. \tag{14}$$

Bracket generation requires $\beta_1 > 0$: if $\beta_1 = 0$, then $\Delta \subset \mathfrak{m}_2 \oplus \mathfrak{m}_3 \subset \mathfrak{so}(2n)$, and since also $\mathfrak{h} = \mathfrak{su}(n) \subset \mathfrak{so}(2n)$, we get $\mathfrak{h} + \Delta + [\Delta, \Delta] \subset \mathfrak{so}(2n) \subsetneq \mathfrak{so}(2n+1)$, so generation fails. Properness then forces $\beta_2 = 0$: indeed, if $\beta_2 > 0$, then (14) gives $\beta_3 = \big((n-1)\beta_2 + \beta_1\big)/n > 0$, so all three coefficients are positive and A is positive definite (full rank), contradicting properness; note that $\beta_3 = 0$ is incompatible with $\beta_1 > 0$, since it would give $\beta_1 = -(n-1)\beta_2 \leq 0$. With $\beta_2 = 0$, (14) gives $\beta_3 = \beta_1/n > 0$, so the boundary structures of this row are exactly $(\beta_1, 0, \beta_1/n)$, $\beta_1 > 0$, as displayed in Table 2. These conditions are also sufficient, by Lemma 3.1.

When $n = 3$, the first two SU(3)-modules $\mathfrak{m}_1$ and $\mathfrak{m}_2$ are equivalent; the GO condition nevertheless forces the displayed splitting; see [6, Sec. 5.2.3].

Case 3: row (9). Here $G/H = \mathrm{SU}(2n+1)/\mathrm{Sp}(n)$, $n \geq 2$, with $\mathfrak{h} = \mathfrak{sp}(n)$ and

$$\mathfrak{m}_1 = \mathfrak{su}(2n+1) \ominus \mathfrak{u}(2n), \qquad \mathfrak{m}_2 = \mathfrak{su}(2n) \ominus \mathfrak{sp}(n), \qquad \mathfrak{m}_3 = \mathfrak{u}(2n) \ominus \mathfrak{su}(2n).$$

The positive coefficients of the source metrics are unrestricted: every positive invariant metric on this space is GO. Bracket generation is equivalent to $\beta_1 > 0$. Indeed, $\mathfrak{h} + \mathfrak{m}_2 + \mathfrak{m}_3 = \mathfrak{u}(2n)$ is a proper subalgebra of $\mathfrak{su}(2n+1)$, and its orthogonal complement is $\mathfrak{m}_1$; if $\beta_1 = 0$, then $\Delta \subset \mathfrak{u}(2n)$ and generation fails, while if $\beta_1 > 0$, then Δ contains $\mathfrak{m}_1$, which generates $\mathfrak{su}(2n+1)$ by the argument of Case 1. Properness is exactly $\beta_2\beta_3 = 0$: if $\beta_2\beta_3 > 0$, then together with $\beta_1 > 0$ all three coefficients are positive and A is full rank; conversely, if $\beta_2\beta_3 = 0$, then A is nonzero (since $\beta_1 > 0$) and rank-deficient, hence proper.

Case 4: row (10). Here $G/H = \mathrm{Spin}(8)/G_2$, and the two irreducible isotropy summands are equivalent seven-dimensional $\mathfrak{g}_2$-modules (they are interchanged by an outer automorphism of $\mathfrak{so}(8)$). For this reason the Riemannian classification [6, Table 1] permits every positive invariant operator, so every nonnegative invariant operator is GO by Lemma 3.1. A proper nonzero one has rank seven, and we now determine its form and its bracket-generation condition explicitly.

Identify $\mathfrak{so}(8) = \mathrm{Der}(\mathbb{O}) \oplus L_{\mathrm{Im}\,\mathbb{O}} \oplus R_{\mathrm{Im}\,\mathbb{O}}$, where $\mathfrak{h} = \mathfrak{g}_2 = \mathrm{Der}(\mathbb{O})$ and L_x, R_x denote left and right multiplication by x on the division algebra of octonions $\mathbb{O}$; the two isotropy summands are $L_{\mathrm{Im}\,\mathbb{O}}$ and $R_{\mathrm{Im}\,\mathbb{O}}$, both isomorphic to the seven-dimensional irreducible $\mathfrak{g}_2$-module. By Schur's lemma, every $\mathfrak{h}$-equivariant self-adjoint endomorphism of this irreducible module is a scalar, so the $\mathfrak{h}$-invariant seven-dimensional subspaces of $L_{\mathrm{Im}\,\mathbb{O}} \oplus R_{\mathrm{Im}\,\mathbb{O}}$ are exactly the graphs

$$V_{a,b} = \{aL_x + bR_x : x \in \mathrm{Im}\,\mathbb{O}\}, \qquad [a : b] \in \mathbb{R}P^1,$$

and a proper nonzero invariant operator is $A = \beta \mathrm{Id}|_{V_{a,b}}$ with $\beta > 0$.

To check bracket generation, write $c = [x, y] = xy - yx$ (the commutator, which lies in $\mathrm{Im}\,\mathbb{O}$) and

$$D_{x,y}(z) = [[x, y], z] - \tfrac{1}{3}\big((xy)z - x(yz)\big);$$

this is an octonion derivation, hence lies in $\mathrm{Der}(\mathbb{O}) = \mathfrak{h}$; see [3, Sec. 4.1]. Using alternativity one computes, for $x, y \in \mathrm{Im}\,\mathbb{O}$,

$$3[aL_x + bR_x, aL_y + bR_y] = (a^2 + 2ab - 2b^2)L_c + (2a^2 - 2ab - b^2)R_c + 2(a^2 + b^2 - ab)D_{x,y}.$$

Hence, modulo $\mathfrak{h}$, the bracket $[V_{a,b}, V_{a,b}]$ is spanned by the vectors $\alpha L_c + \beta R_c$ with c ranging over $\operatorname{Im}\mathbb{O}$, where $\alpha = a^2 + 2ab - 2b^2$ and $\beta = 2a^2 - 2ab - b^2$. Since the commutators $[x, y]$ span $\operatorname{Im}\mathbb{O}$, the spaces $\{aL_x + bR_x\}$ and $\{\alpha L_x + \beta R_x\}$ together span $L_{\operatorname{Im}\mathbb{O}} \oplus R_{\operatorname{Im}\mathbb{O}}$ exactly when the coefficient matrix $\left(\begin{smallmatrix} a & b \\ \alpha & \beta \end{smallmatrix}\right)$ has rank two, that is, when

$$a\beta - b\alpha = a(2a^2 - 2ab - b^2) - b(a^2 + 2ab - 2b^2) = (a + b)(2a - b)(a - 2b) \neq 0.$$

In this case $\mathfrak{h} + V_{a,b} + [V_{a,b}, V_{a,b}] = \mathfrak{so}(8)$ and the structure is bracket generating. If the determinant vanishes, then (α, β) is proportional to (a, b), so $\{\alpha L_x + \beta R_x\} = V_{a,b}$ and $[V_{a,b}, V_{a,b}] \subset \mathfrak{h} \oplus V_{a,b}$; thus $\mathfrak{h} + V_{a,b}$ is already a proper Lie subalgebra and generation fails. This proves the three exclusions $[1 : -1]$, $[1 : 2]$, $[2 : 1]$ of Table 2 and completes the analysis of row (10). □

3.2. **The semisimple case: proof of Theorem 1.2(B).** Here $r \geq 2$, so $\mathfrak{h}$ projects nontrivially onto at least two simple ideals. For $i \in P$, write $\mathfrak{g}_i = \mathfrak{h}_i \oplus \mathfrak{p}_i$, where $\mathfrak{h}_i = \rho_i(\mathfrak{h})$ and the sum is B_i-orthogonal. Identify active copies of $\mathfrak{h}$ using ρ_i. The orthogonal decomposition is

$$\mathfrak{m} = \bigoplus_{i \in P} \mathfrak{p}_i \oplus \Big\{(z_i)_{i \in J} : \sum_{i \in J} z_i = 0\Big\}. \tag{15}$$

The quadratic form of A_c is $c_i B_i$ on the summands $\mathfrak{p}_i$ and

$$q_c(z) = \sum_{i \in J} c_i |z_i|^2, \qquad \sum_{i \in J} z_i = 0,$$

on the diagonal summand, where $|\cdot|$ is induced by $B_{\mathfrak{h}}$. Consequently, on its image, the sub-Riemannian metric is $Q = \big(\sum_i c_i^{-1} B_i\big)\big|_{\operatorname{Im} A_c}$.

Lemma 3.3. *With notation as above, for any coefficients c_i the sub-Riemannian metric defined by A_c is geodesic orbit.*

Proof. For every $p \in \mathfrak{m}$, take $Z = P_{\mathfrak{h}} Cp$. Then

$$A_c p + Z = Cp, \qquad [p, A_c p + Z] = [p, Cp] = 0.$$

Thus every nonnegative cometric is GO whenever its image is bracket generating. □

Lemma 3.4. *With notation as above, a nonnegative GO operator $A : \mathfrak{m} \to \mathfrak{m}$ must be of the form (13).*

Proof. By Lemma 3.1, for every $t > 0$ the Riemannian metric with metric operator $(A + tI)^{-1}$ is G-GO. Hence it is G-naturally reductive and has the form (13), by Theorems 1 and 2 of [6]. □

Lemma 3.5. *Let $V \neq 0$ be a real inner product space and $r \geq 2$. The form $\sum_{i=1}^r c_i |z_i|^2$ on $\sum_i z_i = 0$ is nonnegative if and only if either all $c_i \geq 0$, or exactly one coefficient c_j is negative, every other coefficient is positive, and*

$$c_j \geq -\Big(\sum_{i \neq j} c_i^{-1}\Big)^{-1}.$$

In the first case its kernel consists of vectors supported on the zero coefficients and having coordinate sum zero. In the second case it is positive definite for strict inequality; at equality its kernel is $\{(c_i^{-1} u)_i : u \in V\}$.

Proof. Two negative coefficients, or one negative and one zero coefficient, give a negative value on an opposite pair of nonzero coordinates. In the all-nonnegative case the assertion about the kernel follows term by term. In the remaining case put $S = \sum_{i\neq j} c_i^{-1} > 0$. Subject to $\sum_{i\neq j} z_i = -z_j$, the weighted Cauchy–Schwarz inequality gives

$$\sum_{i\neq j} c_i |z_i|^2 \geq \frac{|z_j|^2}{S},$$

with equality exactly when $z_i = -z_j/(Sc_i)$. Hence nonnegativity is equivalent to $c_j + 1/S \geq 0$, and the equality case gives the stated kernel. For strict inequality, vanishing requires $z_j = 0$ and then every other coordinate zero. □

Theorem 3.6. *Under the preceding hypotheses, the three properties*

$$A_c \geq 0, \qquad \Delta = \operatorname{Im} A_c \neq \mathfrak{m}, \qquad \mathfrak{g} = \mathfrak{h} + \Delta + [\Delta, \Delta]$$

hold simultaneously if and only if the coefficients satisfy, for some $j \in F$,

$$c_i > 0 \ (i \neq j), \qquad c_j = -\Big(\sum_{i\in J\setminus\{j\}} c_i^{-1} \Big)^{-1} < 0, \qquad P \neq \varnothing \text{ or } r \geq 3. \tag{16}$$

Moreover, the kernel is

$$\operatorname{Ker} A_c = \{n(u) : u \in \mathfrak{h}\}, \qquad n(u)_i = c_i^{-1}\rho_i(u), \quad i \in J. \tag{17}$$

Consequently the corank is $\dim \mathfrak{h}$ *and the horizontal rank is* $\dim \mathfrak{g} - 2\dim \mathfrak{h}$.

Proof. Necessity. Nonnegativity and bracket generation force $c_i > 0$ for every $i \in P$. Indeed, a negative coefficient makes A_c negative on the nonzero summand $\mathfrak{p}_i$ in (15). If such a coefficient is zero, the projection of $\mathfrak{h} + \operatorname{Im} A_c$ to $\mathfrak{g}_i$ is contained in the proper subalgebra $\mathfrak{h}_i$; brackets cannot generate $\mathfrak{g}_i$.

If all active coefficients are nonnegative, singularity requires at least two zero coefficients, and their indices must be full. Choose distinct such indices a, b. For every $u \in \mathfrak{h}$, the vector with active coordinates $z_a = u$, $z_b = -u$ and all others zero belongs to $\operatorname{Ker} A_c$. Since A_c is self-adjoint, its image is orthogonal to this kernel, and therefore its a and b coordinates agree under the identifications with $\mathfrak{h}$. The same is true of every element of $\mathfrak{h}$. The equality of these two coordinates defines a proper Lie subalgebra of $\mathfrak{g}$, contradicting bracket generation.

There is consequently one negative active coefficient, at some $j \in F$. Lemma 3.5 implies that every other active coefficient is positive; hence $c_i > 0$ for all $i \neq j$. The strict inequality of that lemma would make A_c positive definite. Singularity therefore forces

$$c_j = -1/S, \qquad S = \sum_{i\in J\setminus\{j\}} c_i^{-1}, \qquad \sum_{i\in J} c_i^{-1} = 0.$$

The kernel is precisely (17).

If $r = 2$ and $P = \varnothing$, both active factors are full and the last summand in (15) is the entire kernel. Thus Δ has no active component. The projection of $\mathfrak{h} + \Delta$ onto the two active factors is diagonal, so it cannot generate their direct sum. Necessity of the final condition in (16) follows.

Sufficiency. Assume (16) holds. Lemma 3.5 proves nonnegativity, singularity and (17). Write $N = \operatorname{Ker} A_c$; since $(\mathfrak{h}+\Delta)^{\perp_B} = N$, it suffices to prove that a vector $n(u) \in N$ orthogonal to $[\Delta,\Delta]$ must vanish.

First suppose $P \neq \varnothing$ and choose $k \in P$. We have $\mathfrak{p}_k \subset \Delta$. The span of the $\mathfrak{h}_k$-projections of $[\mathfrak{p}_k,\mathfrak{p}_k]$ is an $\mathfrak{h}_k$-ideal, by equivariance. It is nonzero: otherwise invariance of B_k would give

$$B_k([U,X],Y) = B_k(U,[X,Y]) = 0 \qquad (U \in \mathfrak{h}_k,\ X,Y \in \mathfrak{p}_k).$$

Because $[\mathfrak{h}_k,\mathfrak{p}_k] \subset \mathfrak{p}_k$, this would imply $[\mathfrak{h}_k,\mathfrak{p}_k] = 0$ and make $\mathfrak{h}_k$ a nonzero proper ideal of the simple $\mathfrak{g}_k$. The span is therefore all of $\mathfrak{h}_k$. Orthogonality of $n(u)$ to these brackets says

$$0 = B(n(u),[X,Y]) = c_k^{-1} B_k(\rho_k(u),[X,Y]) \qquad (X,Y \in \mathfrak{p}_k),$$

which forces $u = 0$.

Now suppose $P = \varnothing$, so $r \geq 3$ and all active factors are full. Put $v_i = c_i^{-1}$ for $i \in F$ and choose a nonzero real vector

$$a \in W = \Big\{(a_i)_{i\in F} : \sum_i a_i = 0,\ \sum_i v_i a_i = 0\Big\}.$$

This is possible because $\dim W = r-2$; the vectors $(1)_i$ and $(v_i)_i$ are independent. For every $x \in \mathfrak{h}$, the vector $a \otimes x := (a_i \rho_i(x))_{i\in F}$ lies in $\Delta = \mathfrak{m} \cap N^{\perp_B}$. The identities $\sum_i v_i = \sum_i v_i a_i = 0$ give

$$\kappa := \sum_i v_i a_i^2 = \sum_{i\neq j} v_i (a_i - a_j)^2 > 0.$$

Strict positivity follows because all v_i for $i \neq j$ are positive; equality would make all a_i equal, and $\sum_i a_i = 0$ would then imply $a = 0$. For $x,y \in \mathfrak{h}$,

$$B\big(n(u),[a\otimes x, a\otimes y]\big) = \kappa B_{\mathfrak{h}}(u,[x,y]).$$

Since $[\mathfrak{h},\mathfrak{h}] = \mathfrak{h}$, orthogonality forces $u = 0$. This proves $\mathfrak{h} + \Delta + [\Delta,\Delta] = \mathfrak{g}$ in both cases. The metric is positive definite on the image, and formula (17) gives the rank and corank. □

Theorem 1.2(B) is indeed the following theorem.

Theorem 3.7. *Under the above hypotheses. Let $(G/H, \Delta, \langle\cdot,\cdot\rangle_\Delta)$ be a compact, connected, simply connected sub-Riemannian geodesic orbit manifold with G semisimple and H simple. Then it is isometric to $((K\times H)/\operatorname{diag} H, \Delta, B|_\Delta) = (K,\Delta,b)$, where K is a compact simply connected semisimple Lie group, b is a positive bi-invariant metric on K, and $\Delta = \mathfrak{h}^{\perp_b} \subset \mathfrak{k}$ is the b-orthogonal complement of the Lie algebra $\mathfrak{h}$ of H in the Lie algebra $\mathfrak{k}$ of K, $H \simeq (\rho_i(H))_{i\neq j} \subset K$ is a connected simple subgroup that is not normal.*

Proof. Notation as in Theorem 3.6. Put $\mathfrak{k} = \bigoplus_{i\neq j} \mathfrak{g}_i$ and $b = \sum_{i\neq j} c_i^{-1} B_i$. The remaining projection embeds $\mathfrak{h}$ into $\mathfrak{k}$. At the group level, replace H by its image in K, dividing out the possible finite ineffective kernel. The homogeneous quotient $(K\times H)/\operatorname{diag} H$ is K, with tangent map

$$T\big((x_i)_{i\neq j}, z_j\big) = \big(x_i - \rho_i(z_j)\big)_{i\neq j},$$

where ρ_i is the isotropy embedding. For $p \in \mathfrak{m}$ with active components z_i, orthogonality to $\operatorname{Ker} A_c$ reads

$$\sum_{i\neq j} z_i/c_i - Sz_j = 0.$$

This is precisely $b(Tp, \mathfrak{h}) = 0$. Also $T(A_c p)_i = c_i p_i - c_j \rho_i(z_j)$, and $\sum_{i\neq j} z_i = -z_j$, $c_j S = -1$ give

$$b\big(T(A_c p), T(A_c p)\big) = \sum_{i\neq j} c_i |p_i|^2 + (2c_j + c_j^2 S)|z_j|^2 = B(p, A_c p).$$

Polarization proves that the resulting horizontal metric is exactly $Q = b|_\Delta$.

It remains to determine bracket generation. The Lie algebra generated by $\mathfrak{h}^{\perp_b}$ is $\mathfrak{h}$-invariant, hence is an ideal of $\mathfrak{k} = \mathfrak{h} + \mathfrak{h}^{\perp_b}$. Its b-orthogonal complement is also an ideal and lies in $\mathfrak{h}$. Simplicity of $\mathfrak{h}$ makes this complement either zero or $\mathfrak{h}$. The latter occurs exactly when $\mathfrak{h}$ is an ideal of $\mathfrak{k}$. This proves the stated condition that H be non-normal. □

Example 3.8. Let $\mathfrak{g} = \mathfrak{h} \oplus \mathfrak{h} \oplus \mathfrak{h}$, with diagonally embedded compact simple $\mathfrak{h}$, equal normalized forms, and $(c_1, c_2, c_3) = (1, 1, -\frac{1}{2})$. Then

$$\operatorname{Ker} A_c = \{(u, u, -2u) : u \in \mathfrak{h}\}, \qquad \Delta = \{(x, -x, 0) : x \in \mathfrak{h}\}.$$

On Δ, $A_c = I$ and $Q = B|_\Delta$. Brackets of horizontal vectors have the form $([x, y], [x, y], 0)$; together with Δ and the diagonal isotropy they span all of $\mathfrak{g}$.

Example 3.9. Let $\rho : \mathfrak{h} \hookrightarrow \mathfrak{g}_1$ be a proper embedding into a compact simple algebra, and set $\mathfrak{g} = \mathfrak{g}_1 \oplus \mathfrak{h}$ with isotropy $\{(\rho(u), u) : u \in \mathfrak{h}\}$. With normalized forms, take $(c_1, c_2) = (a, -a)$, $a > 0$. Then $\Delta = \mathfrak{p}_1 \oplus 0$, and $Q = a^{-1} B_1$ on this subspace. This is an admissible two-active-factor case because the first projection is proper. If instead $\mathfrak{g}_1 = \mathfrak{h}$ and ρ is an isomorphism, then $A_c = 0$ on the active tangent space and bracket generation fails.

Declarations

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